

Homework (Habanero)

- Q1.** A hospital uses a ratio of 3 : 5 for saline to water in IV solutions. If 18 ounces of saline are used, how many ounces of water are needed?
(A) 10
(B) 12
(C) 15
(D) 20
(E) 30
- Q2.** A biologist studies the growth rate of cells. The ratio of healthy cells to infected cells is 2 : 3. If there are 12 healthy cells, how many infected cells are there?
(A) 12
(B) 15
(C) 18
(D) 20
(E) 24
- Q3.** A patient needs a 1 : 4 ratio of medication to water. If 6 mL of medication are used, how many mL of water are required?
(A) 12
(B) 16
(C) 20
(D) 24
(E) 30
- Q4.** A medical study analyzes the ratio of patients with a certain disease to those without it, which is 1 : 9. If 45 patients without the disease are studied, how many patients with the disease are there?
(A) 3
(B) 5
(C) 6
(D) 9
(E) 12
- Q5.** The ratio of red blood cells to white blood cells in a healthy individual is 7 : 1. If there are 21 red blood cells, how many white blood cells are there?
(A) 2
(B) 3
(C) 5
(D) 7
(E) 10
- Q6.** A researcher examines the growth of bacteria in a lab. The ratio of bacteria to their food source is 3 : 2. If there are 18 bacteria, how many units of food are needed?
(A) 8
(B) 10
(C) 12
(D) 15
(E) 20
- Q7.** The dosage of a certain medication is 2 : 5 for adults to children. If an adult receives 10 mL of the medication, how many mL does a child receive?
(A) 4
(B) 5
(C) 6
(D) 8
(E) 10
- Q8.** A hospital pharmacy uses a ratio of 4 : 7 for generic to name-brand medications. If 28 generic medications are dispensed, how many name-brand medications are dispensed?
(A) 35
(B) 40
(C) 45
(D) 49
(E) 56

Open Ended Questions (FRQ)

- Q9.** A doctor prescribes a ratio of 1 : 3 for a certain medication and a vitamin supplement. If a patient takes 12 mL of the medication, how many mL of the vitamin supplement should they take, and what is the total volume of the mixture?
- Q10.** A researcher studies the effect of a certain drug on the growth of cancer cells. The ratio of cancer cells to healthy cells is 2 : 5. If there are 50 cancer cells, how many healthy cells are there, and what is the percentage of cancer cells?

Chef's Tips

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- Chef's Tips: Ensure students use correct units and dimensional analysis.
 - Chef's Tips: Remind students to check their work with proportions.
 - Chef's Tips: Encourage students to use real-world examples to demonstrate understanding.